Charlone Gianne V. Cruz

Full-Stack Developer · Machine Learning Researcher · AI Systems Architect

Cebu Institute of Technology – University (CIT-U) | 3rd Year BS Computer Science

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1. Executive Summary & Profile

Charlone Gianne V. Cruz is a 3rd-year Computer Science undergraduate at Cebu Institute of Technology – University (CIT-U) specializing in Applied Machine Learning, Biomedical Time-Series AI, Computer Vision, and Full-Stack Engineering. Charlone bridges rigorous mathematical and algorithmic modeling with robust, production-grade software architecture. Rather than treating machine learning models as isolated scripts, he architects and deploys end-to-end intelligent systems: from dual-model gradient-boosted decision trees predicting maritime fisheries to 1D Temporal Convolutional Neural Networks (1D CNNs) analyzing continuous physiological smartwatch telemetry, to scalable enterprise web backends in Python Django and modern reactive frontends in C# Blazor WebAssembly.

Recently recognized as a **Top 25 Regional Finalist** and winner of the **Best Logo Design Award** at the DICT Philippine Startup Challenge (PSC) Region VII for *PAROLA* (Team Tiki), Charlone actively competes in regional hackathons including the **AIFEST AI Hackathon 2026** and holds verified credentials from the ASEAN Foundation & Google.org and PSITE Central Visayas.

2. Technical Skills Matrix

Domain	Core Technologies, Frameworks & Libraries
Machine Learning & AI	LightGBM, PyTorch, TensorFlow, Scikit-learn, OpenCV, 1D Temporal CNN (TCN), Multi-Layer Perceptrons (MLP), RAG, Classical Machine Learning
Data Science & Biomedical	Pandas, NumPy, Time-Series Feature Engineering, PhysioCGM Biomedical Telemetry Analysis, Data Augmentation & Cleaning, Statistical Visualization
Programming Languages	Python, C#, Java, C, C++, PHP, JavaScript (ES6+), SQL, HTML5, CSS3
Web & App Frameworks	Python Django (MVT Architecture), C# Blazor WebAssembly, React Native, JavaFX, Java Swing, Streamlit
Databases & Tools	SQLite, PostgreSQL, MySQL, Apache, Git, GitHub, REST APIs, Postman, Linux/macOS Shell

3. Ranked Project Portfolio (Prioritized by Algorithmic Depth & Real-World Utility)

Rank 1: PAROLA — Intelligent Marine Catch Prediction System

Philippine Startup Challenge (Region VII Top 25 Finalist & Best Logo Design Winner) | Lead ML Engineer (Team Tiki) | Python, LightGBM, Pandas, Geospatial Data | github.com/Gianne818/Parola_PSC

- Architected dual specialized gradient-boosted decision tree models using LightGBM, training distinct predictive pipelines for demersal (bottom-dwelling) and pelagic (open-water) fish species.
- Modeled sea-surface temperature, ocean currents, and seasonal coordinate shifts to forecast optimal sustainable fishing zones for artisanal fishermen in Region VII.

Rank 2: Nocturnal Hypoglycemia Detection via Wearable CGM Telemetry

Applied AI & Biomedical Deep Learning Research | Lead AI Researcher & Data Analyst | Python, PyTorch, 1D Temporal CNN, MLP, PhysioCGM Dataset

- Conducted rigorous exploratory analysis and data augmentation on the PhysioCGM dataset to assess nocturnal hypoglycemia detection via non-invasive consumer smartwatches.
- Engineered a sequential 1D Temporal Convolutional Neural Network (1D CNN) capturing sequential physiological trends preceding nocturnal glycemic drops, outperforming baseline MLP models.

Rank 3: IMentia — AI Facial Recognition & Memory Recall Assistant

Assistive Healthcare & Computer Vision | Computer Vision & AI Developer | Java, Java Swing, OpenCV (Java Interface), SQLite

- Engineered an assistive desktop platform for Alzheimer's and dementia patients using webcam video feeds and OpenCV facial recognition.
- Automatically surfaces interpersonal relationship logs, past conversation summaries, and memory prompts upon detecting recognized faces.

Rank 4: SpeechToCode — AI Voice-to-Code Synthesis Engine

Developer Accessibility & Natural Language Processing | AI Developer & NLP Engineer | Python, SpeechRecognition, AST Parsing

- Constructed a voice-driven coding assistant translating natural spoken language into syntactically valid source code with error-tolerant phonetic parsing for programming syntax.

Rank 5: LantaWires — Scalable Web Architecture Platform

Full-Stack Web Systems & Backend Engineering | Full-Stack Backend Developer | Python, Django (MVT), SQLite/PostgreSQL, JavaScript | github.com/Gianne818/LantaWires

- Engineered a modular enterprise web platform adopting Django Model-View-Template architecture with secure authentication, role-based controls, and administrative dashboards.

Rank 6: CineTerra — Cinematic Discovery & Media Platform

Modern Web Applications (Single-Page App) | Frontend Engineer | C#, .NET, Blazor WebAssembly, Modern CSS | github.com/Gianne818/CineTerra

- Constructed a high-performance single-page media exploration interface executing client-side C# directly in the browser via Blazor WebAssembly with responsive filtering.

Rank 7: OurDriedFish — Local Artisanal E-Commerce Platform

Enterprise Web Frontend & Regional Digitalization | Frontend Developer | C#, Blazor WebAssembly, Responsive Design | github.com/Gianne818/OurDriedFish

- Designed a modern digital marketplace championing local Cebuano dried seafood producers with reactive component state and interactive catalog browsing.

Rank 8: +inHealth — Desktop Ergonomic Wellness & Activity Monitor

Desktop Software Engineering & Ergonomics | Software Developer | Java, JavaFX, Desktop Event Hooks

- Created an unobtrusive desktop wellness assistant combating programmer screen fatigue and sedentary habits through scheduled physical activity challenges.

Rank 9: WildGuard — Campus Security & Incident Management System

Campus Infrastructure & Information Systems | Backend Developer | PHP, MySQL, Apache, Web Forms

- Built a centralized incident reporting, visitor logging, and physical access management web system for campus security operations.

Rank 10: Sugbuanon Saga: Banhwa — Cultural Mythological Adventure

Game Development & Visayan Folklore | Game Logic & Narrative Developer | Game Jam 2025 Submission

- Developed a story-driven adventure game celebrating Cebuano mythology where players reforge an ancient shattered relic to protect Cebu from mythical monsters.

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4. Honors, Awards & Verified Certifications

- **Best Logo Design Award** — Philippine Startup Challenge (2026), DICT Region VII (Oct 24, 2026, Bai Hotel, Mandaue City). Project: *PAROLA* (Team Tiki).
- **Top 25 Regional Finalist** — Philippine Startup Challenge (2026), DICT Region VII (Oct 24, 2026, Bai Hotel, Mandaue City). Project: *PAROLA* (Team Tiki).
- **AIFEST AI Hackathon 2026** — Professional Competency Credentials: *Workshop 1 (Human-Centered AI Design)*, *Workshop 2 (Technical System Architecture)*, and *Webinar 3 (Agile Pitching & Venture Strategy)*.
- **AI Ready ASEAN Certificate of Completion** — ASEAN Foundation & Google.org (Dec 13, 2025). AI Fundamentals, Ethics, and Practical Application.
- **Cebu ICT Student Congress 2026 Certificate of Participation** — PSITE Central Visayas (April 30, 2026, UC New Cebu Coliseum). Emerging AI and Cloud Technologies.
- **Game Jam 2025 Submission** — Official Game Developer for *Sugbuanon Saga: Banhwa*.

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5. Academic Background

Cebu Institute of Technology – University (CIT-U), Cebu City, Philippines
Bachelor of Science in Computer Science (BSCS) — 3rd Year Undergraduate (Age 20) | Relevant Course: CSIT 349 (Applied AI)